

Catalyst Sentiment Prediction

1. Project Title and Team Members

Team Members:

- Camela Phung
- Nathan Nguyen

2. Project Description

This is a model designed to predict sentiment of financial texts (news articles, and earnings call transcripts) as bullish, bearish, or neutral and its impact to the corresponding stock or index.

Our project aims to develop a model that can:

- Process and analyze a single text data such as articles, earnings calls, etc. with pretrained NLP model.
- Identify key sentiment indicators and financial terminology
- Classify texts according to the potential impact

This project aims to address the challenge of information overload in financial markets and support traders in making fast decisions during short-term/intraday trading. This model can also act as a part of an algorithmic trading pipeline due to its capability to quickly analyze and output sentiments which can be used as one of the data points.

3. Roles and Contributions

Nathan Nguyen	Camela Phung
Scrap data from relevant news source and company's earning calls	Model Evaluation
Clean and normalize dataset	Find NLP model for text processing
Model Development	
Feature Engineering	
Create slides and report	

4. Course-Related Topics to be Explored

Our project will explore several key topics:

- **Natural Language Processing (NLP):** text processing techniques including tokenization, named entity recognition, and sentiment analysis and NLP models such as BERT/finBERT
- **Classification Model:** Implementation of suitable classification algorithms
- **Feature Engineering:** Establish of meaningful features from text data for financial sentiment analysis
- **Model Evaluation:** Set up appropriate metrics for classification tasks and text analysis

5. Dataset Description

Primary Data Sources:

- Financial news articles (articles from YahooFinance, Bloomberg, CNBC) for one stock or index
- Earnings call transcripts

Dataset Characteristics:

- Time period: January 2020 – March 2025
- Text data paired with corresponding market movements (stock price changes 1-day, 3-day, and 1-week after publication)
- Labeled data for supervised learning (bullish/bearish/neutral classifications)
- Metadata including publication timestamp, source, company/sector mentions, and author

Data Processing:

- Text cleaning and normalization
- Financial entity extraction (company names, financial metrics, etc.)
- Historical market data alignment for ground truth labeling

6. Project Timeline

Week 1: Data Collection and Preprocessing (March 15-21)

- Create a dataset by gathering data from multiple sources
- Clean and normalize dataset

Week 2: Feature Engineering (March 22-28)

- Extract relevant features from text data
- Find a suitable pretrained NLP models to analyze financial texts
- Prepare training and validation datasets

Week 3-4: Model Development (March 29-April 11)

- Develop model for classifying sentiment and predict the impact
- Train and test the model

Week 5: Evaluation and Final Integration (April 12-18)

- Evaluate model for correctness and performance
- Error analysis and model refinement if needed
- Implement an example application to show how the model can be used

Week 6: Documentation and Presentation (April 19-May 1)

- Finalize documentation
- Prepare presentation slides

7. Evaluation Plan

Technical Evaluation:

- **Classification Metrics:** Precision, recall, F1-score, and accuracy for sentiment classification
- **Financial Relevance:** Correlation between predicted sentiment and subsequent market movements
- **Cross-Source Performance:** Comparative analysis of model performance across different text sources (e.g.: news articles vs. earnings calls)

Project Success Criteria:

- Achieve minimum 75% accuracy in bullish/bearish/neutral classification
- Able to correctly demonstrate the correlation between the news and the stock
- Able to process and analyze at least 90% of input texts
- Have a visualization system for the results